## Process structure

10 candidate structures

Chilling  Forcing

Forcing only

2 structures

**Sequential**

4 structures

Parallel

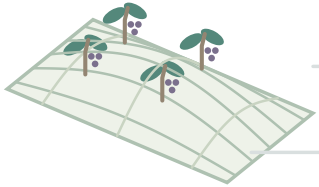
4 structures

**Selected: CB-FL-S**

Beta chilling → linear forcing

## Further improvement

### Terrain-adjusted temperature



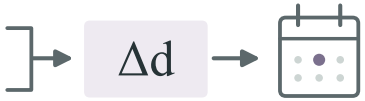
Elevation offset

Recalibration

### Process-guided residual learning

Process states

Environment

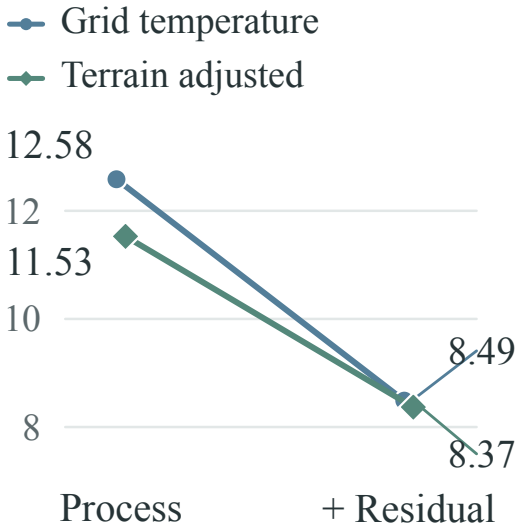


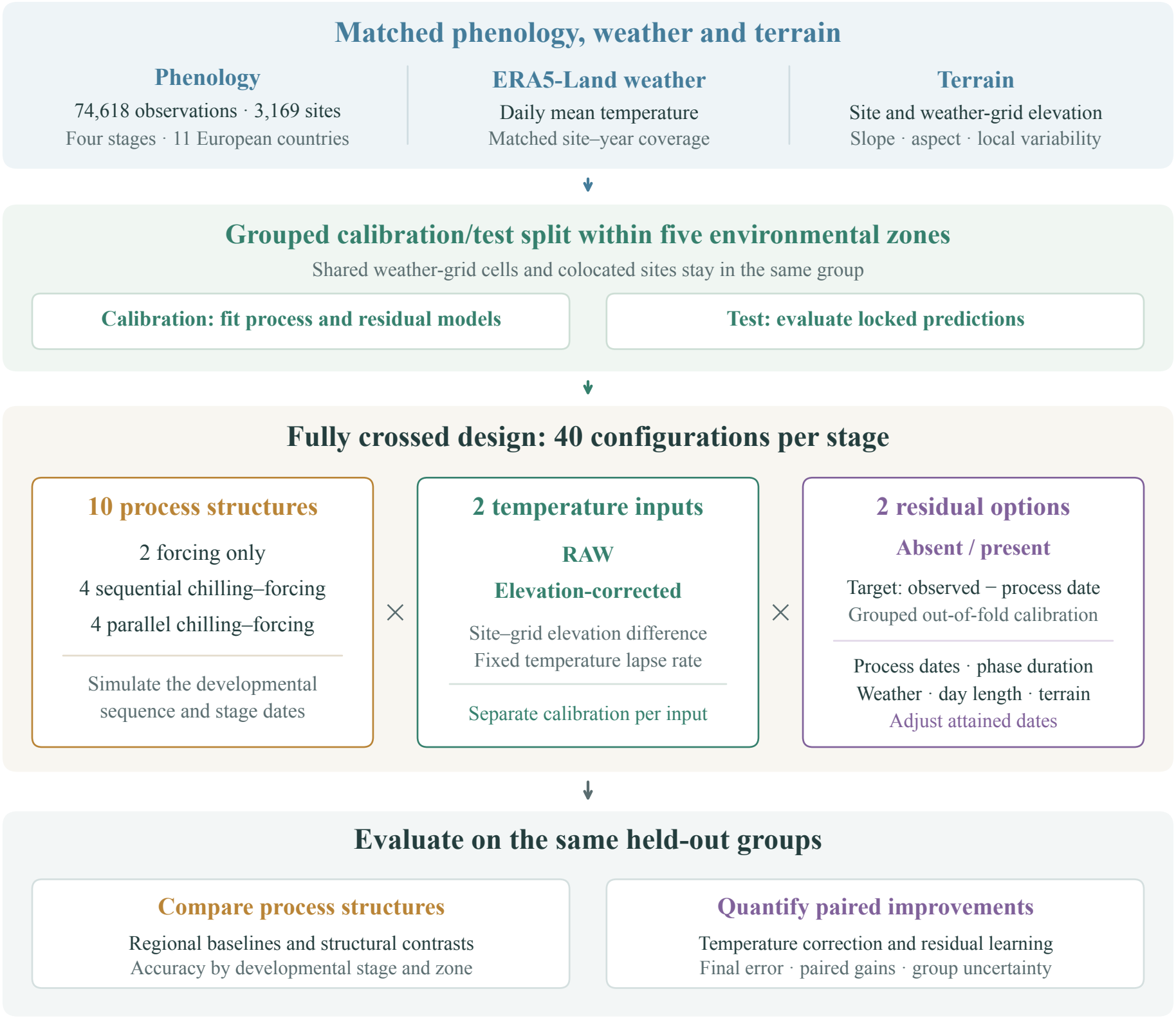

Grouped out-of-fold errors

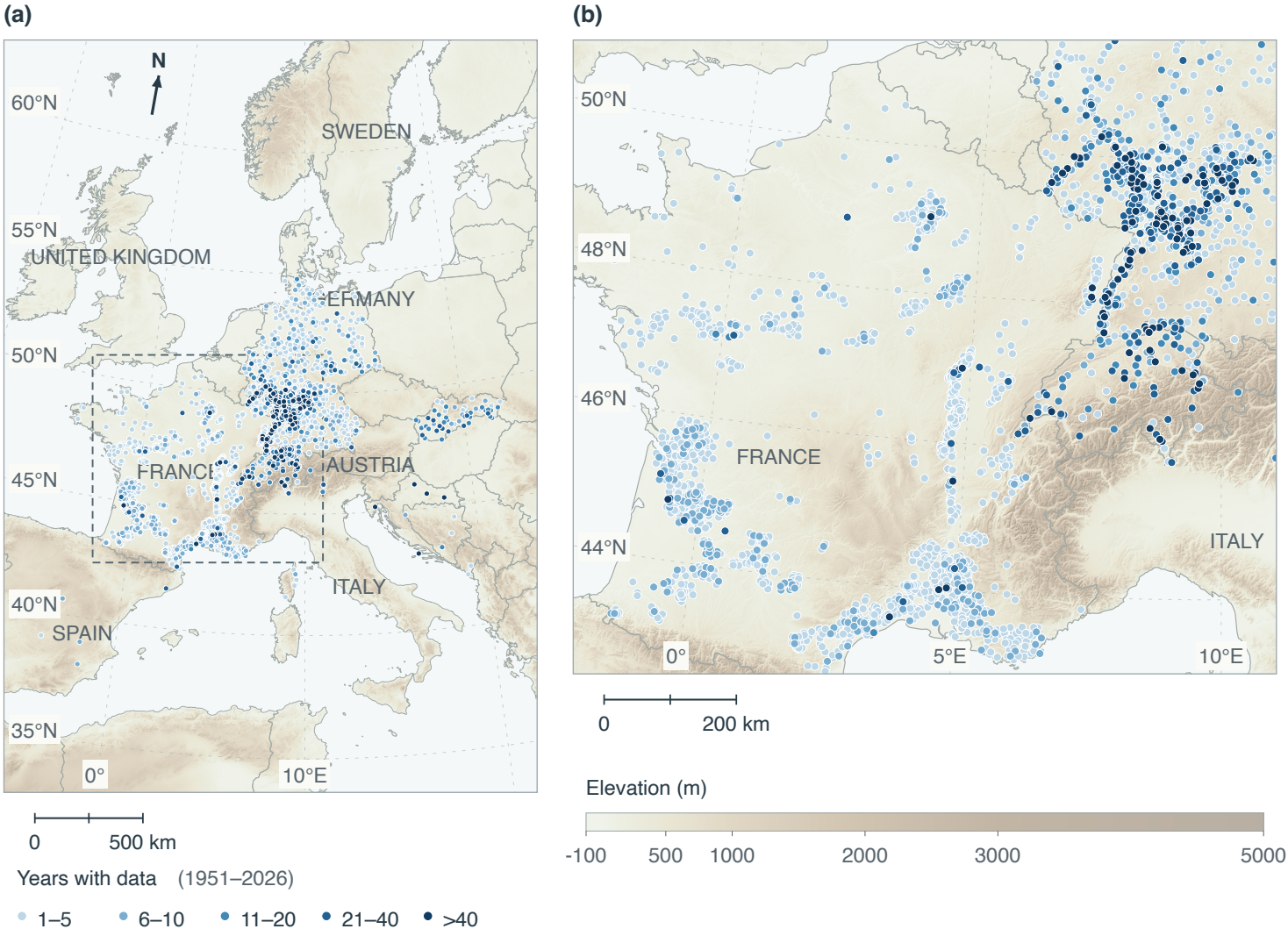
Both corrections evaluated across all 10

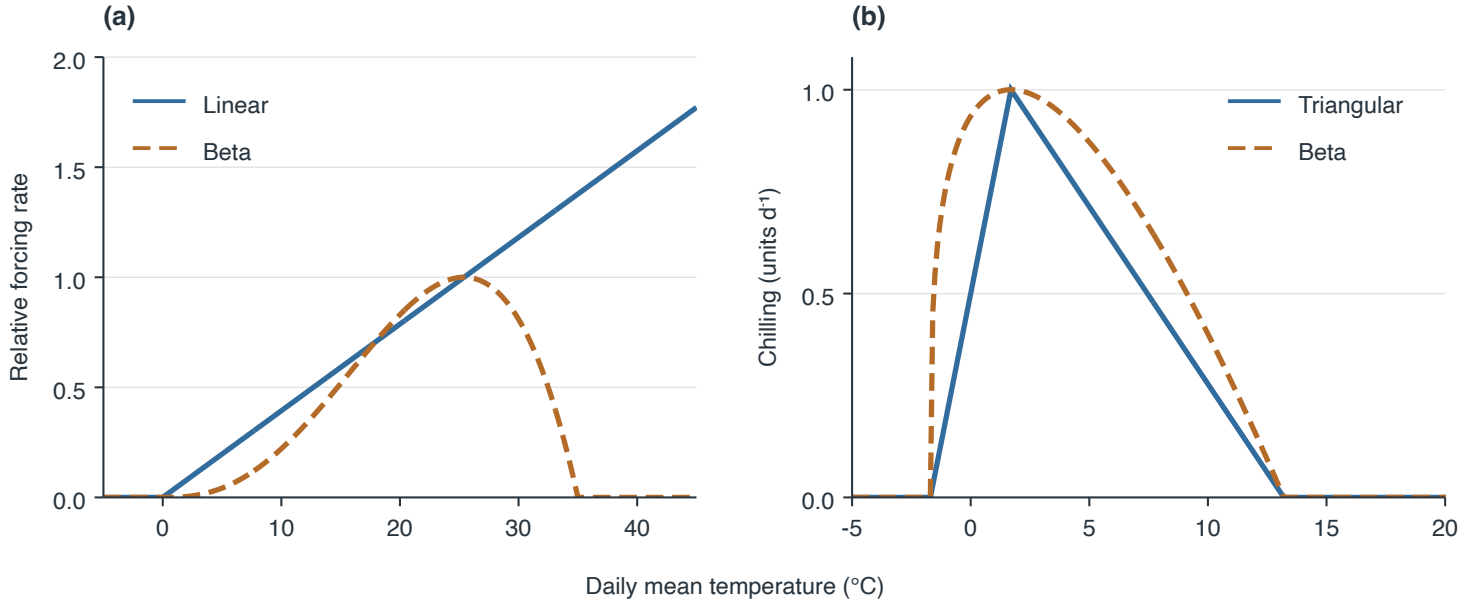
## Grapevine phenology

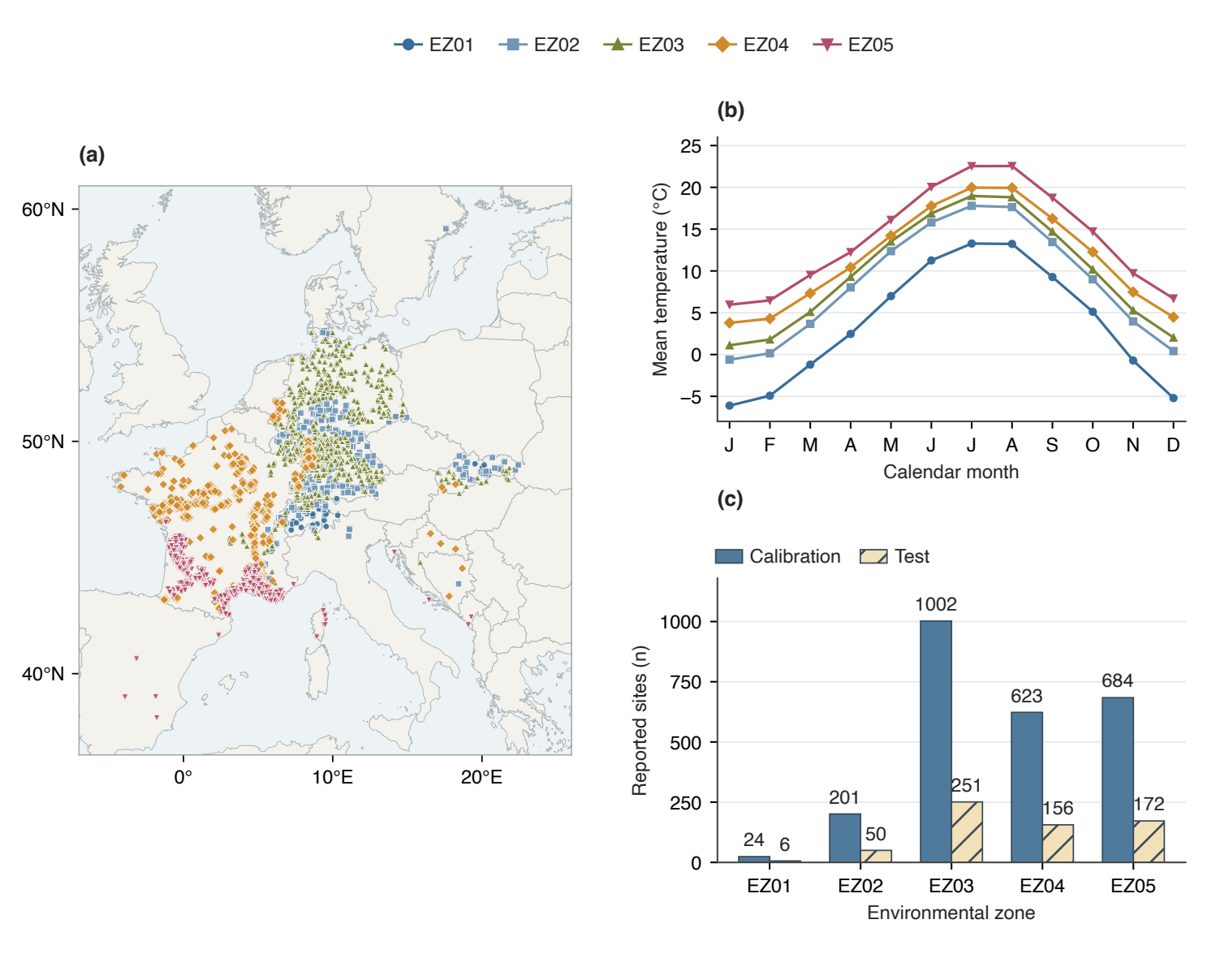
Regional test MAE (days)

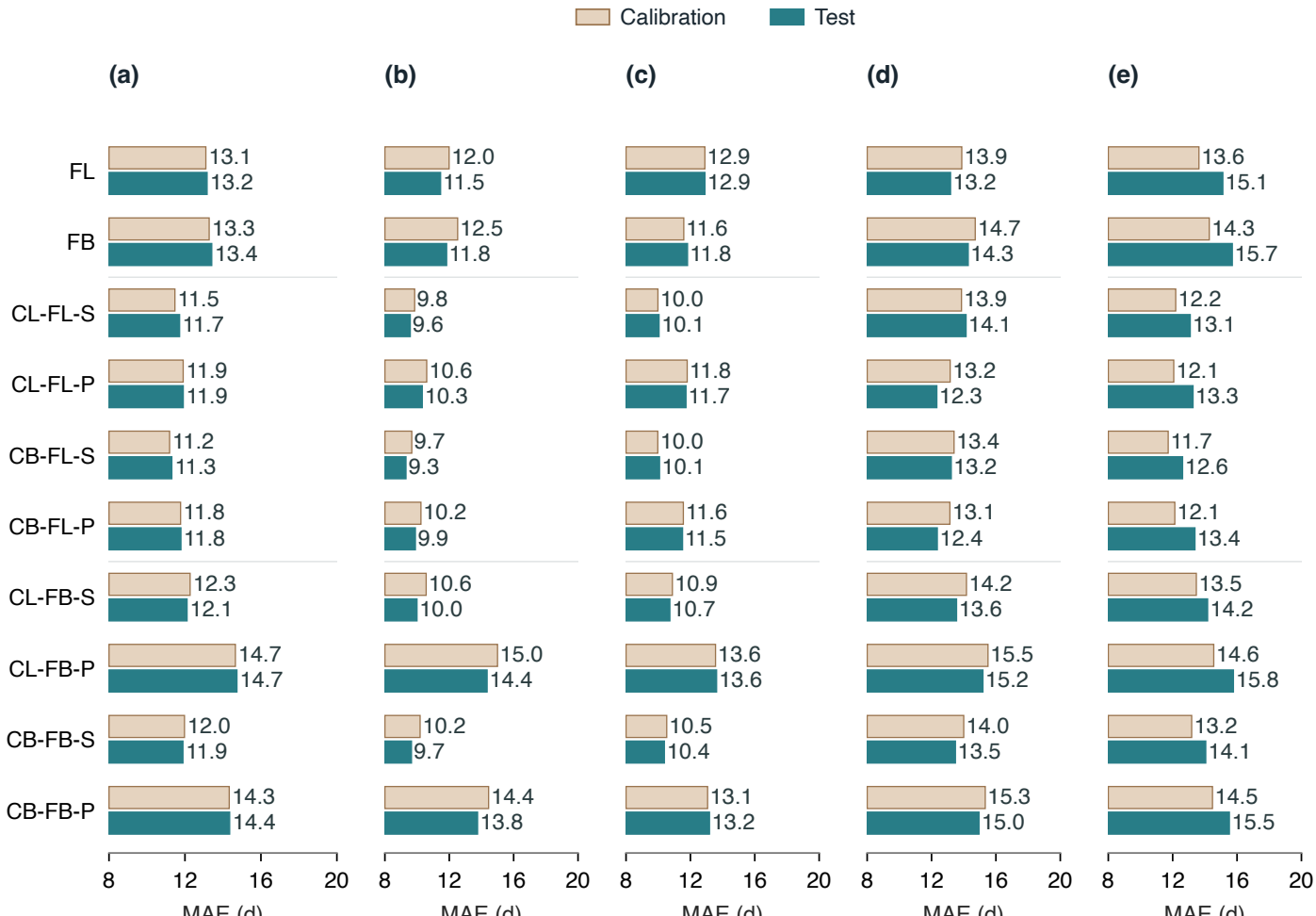


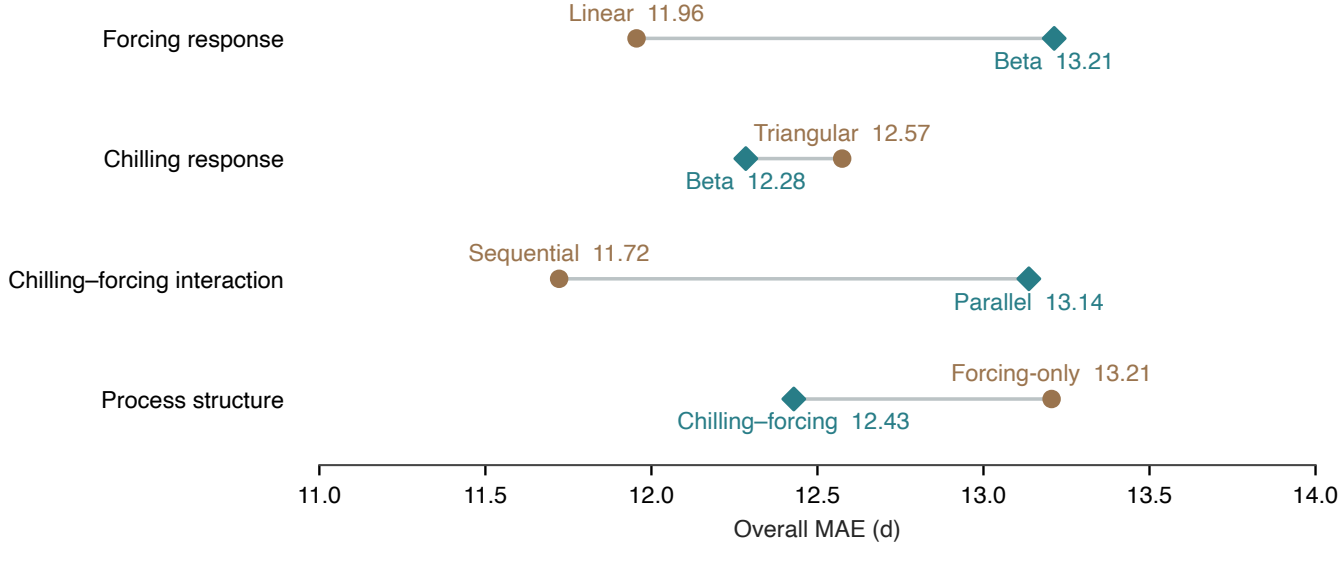

Matched phenology, weather and terrain
Phenology
74,618 observations · 3,169 sites
Four stages · 11 European countries
ERA5-Land weather
Daily mean temperature
Matched site–year coverage
Terrain
Site and weather-grid elevation
Slope · aspect · local variability
Grouped calibration/test split within five environmental zones
Shared weather-grid cells and colocated sites stay in the same group
Calibration: fit process and residual models
Test: evaluate locked predictions
Fully crossed design: 40 configurations per stage
10 process structures
2 forcing only
4 sequential chilling–forcing
4 parallel chilling–forcing
Simulate the developmental sequence and stage dates
×
2 temperature inputs
RAW
Elevation-corrected
Site–grid elevation difference
Fixed temperature lapse rate
Separate calibration per input
×
2 residual options
Absent / present
Target: observed − process date
Grouped out-of-fold calibration
Process dates · phase duration
Weather · day length · terrain
Adjust attained dates
Evaluate on the same held-out groups
Compare process structures
Regional baselines and structural contrasts
Accuracy by developmental stage and zone
Quantify paired improvements
Temperature correction and residual learning
Final error · paired gains · group uncertainty

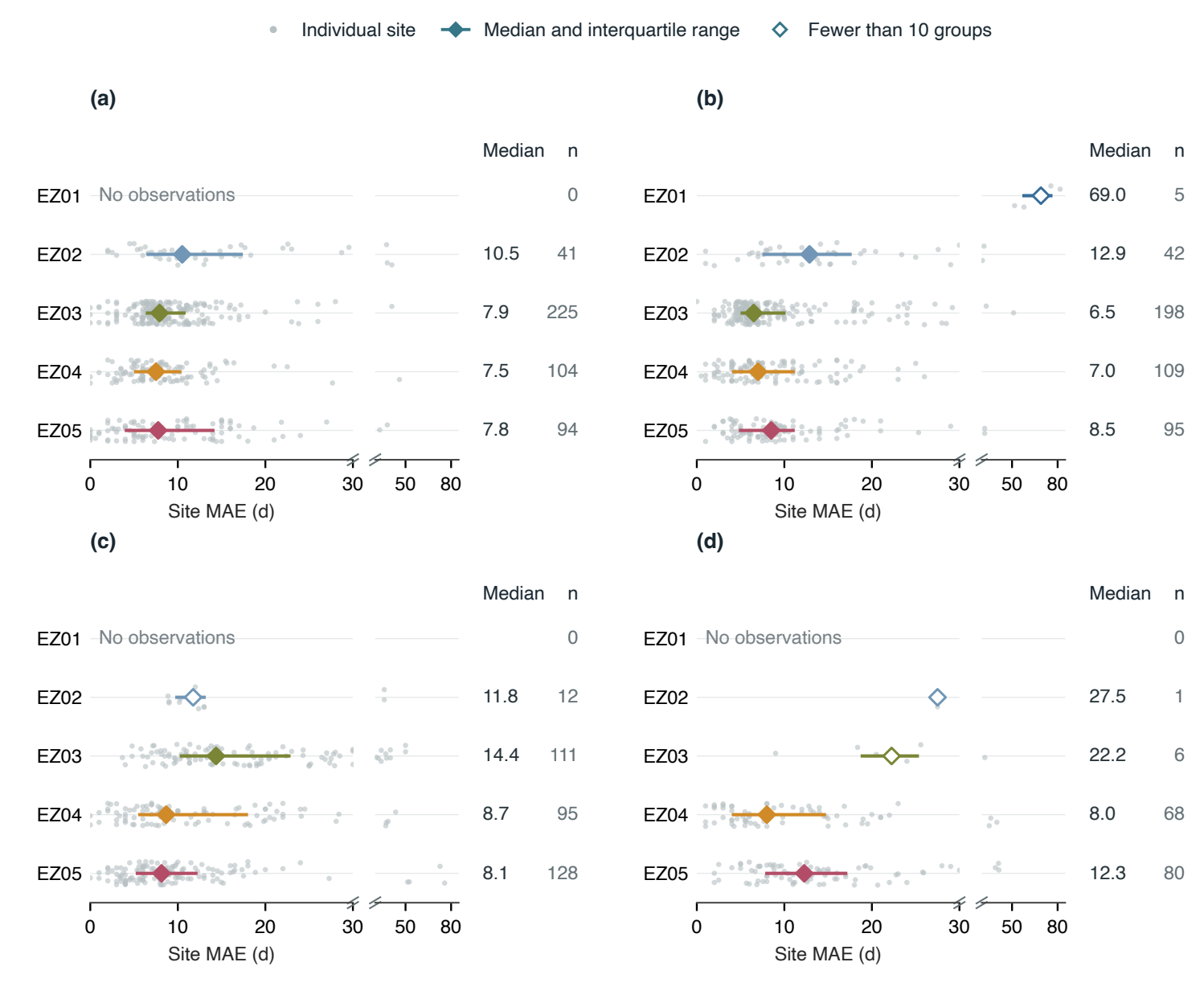

(a)
(b)
N
60°N
55°N
50°N
45°N
40°N
35°N
0°
10°E
SWEDEN
UNITED KINGDOM
GERMANY
FRANCE
AUSTRIA
ITALY
SPAIN
0
500 km
Years with data (1951–2026)
1–5
6–10
11–20
21–40
>40
50°N
48°N
46°N
44°N
0°
5°E
10°E
FRANCE
ITALY
0
200 km
Elevation (m)
-100
500
1000
2000
3000
5000

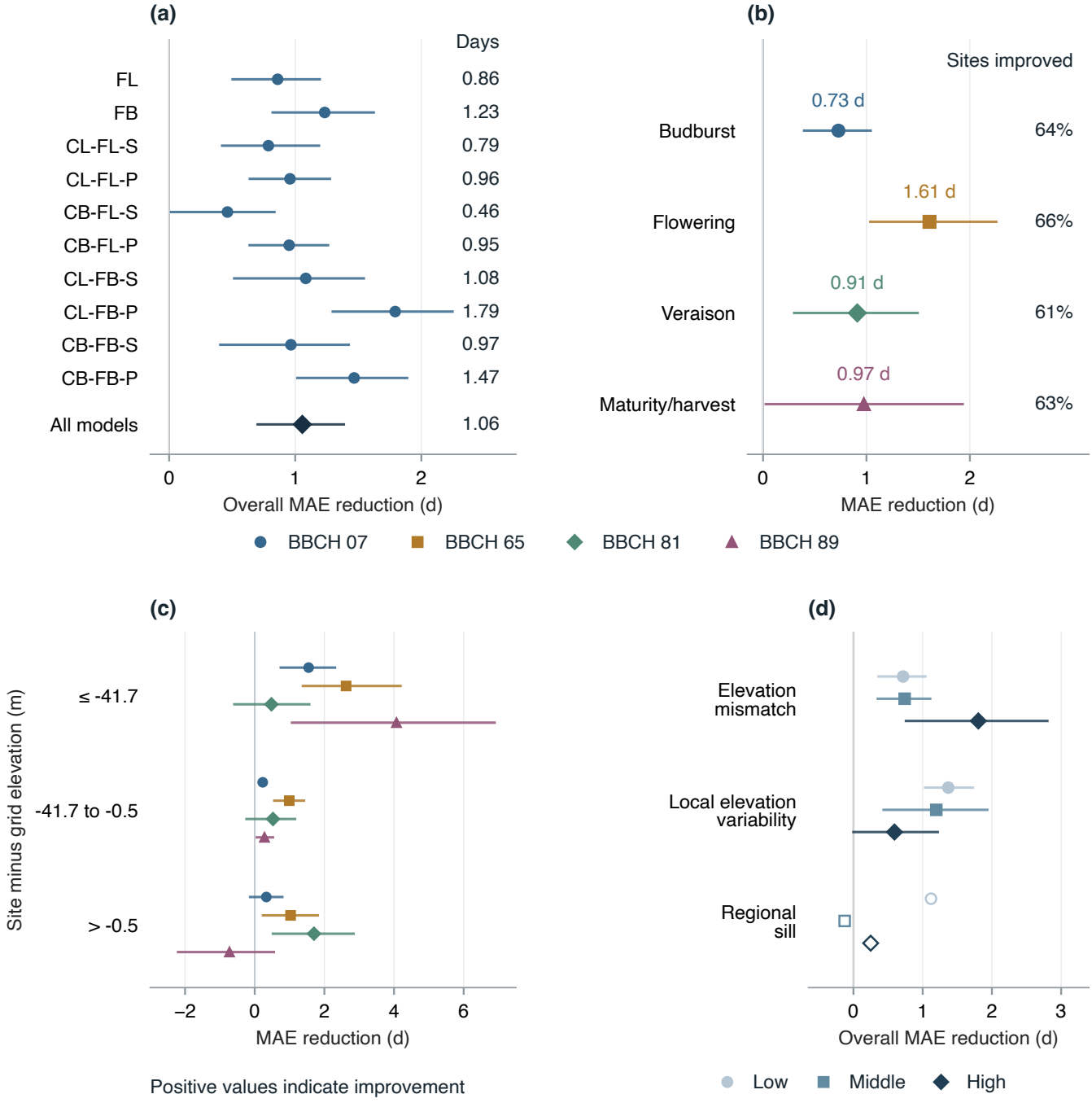

(a)
Linear
Beta
Relative forcing rate
2.0
1.5
1.0
0.5
0.0
0
10
20
30
40
(b)
Triangular
Beta
Chilling (units d-1)
1.0
0.5
0.0
-5
0
5
10
15
20
Daily mean temperature (°C)

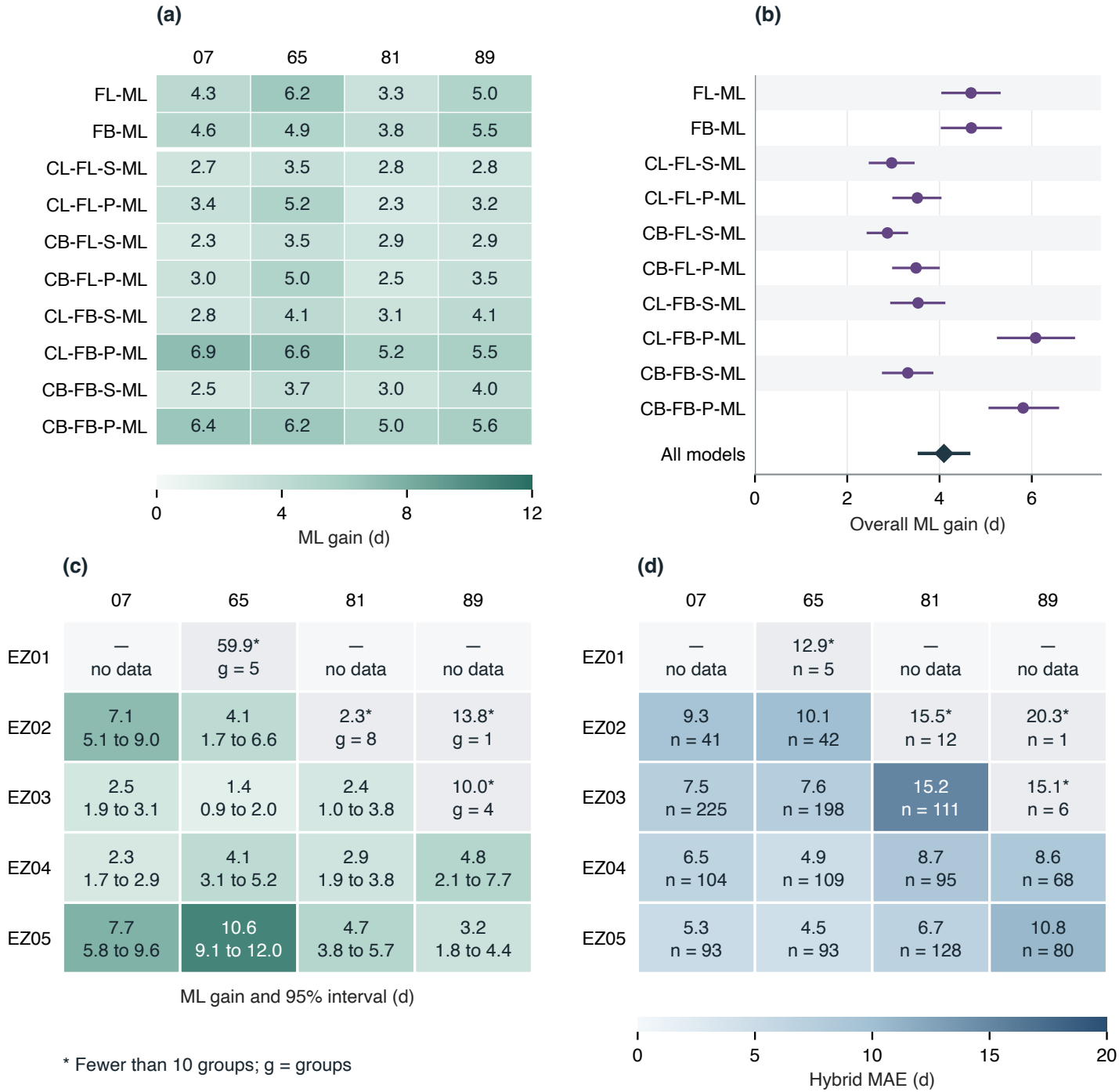

EZ01
EZ02
EZ03
EZ04
EZ05
(a)
60°N
50°N
40°N
0°
10°E
20°E
(b)
Mean temperature (°C)
25
20
15
10
5
0
−5
J
F
M
A
M
J
J
A
S
O
N
D
Calendar month
(c)
Calibration
Test
Reported sites (n)
1000
750
500
250
0
24
6
201
50
1002
251
623
156
684
172
EZ01
EZ02
EZ03
EZ04
EZ05
Environmental zone

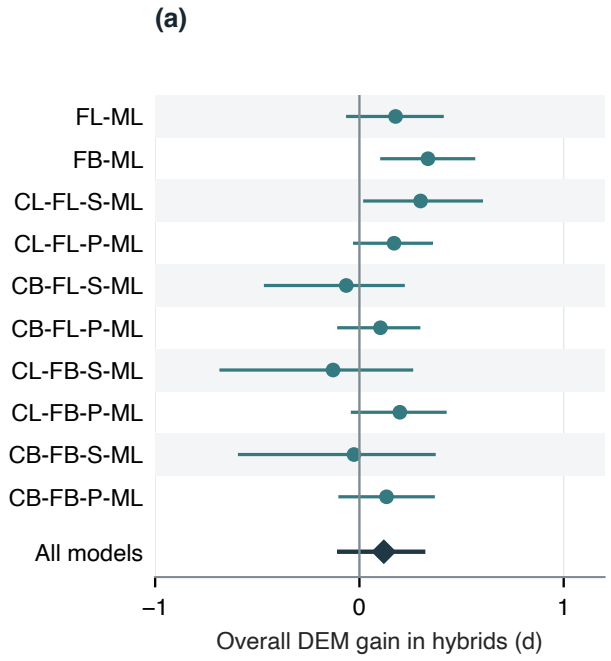

Calibration
Test
(a)
(b)
(c)
(d)
(e)
FL
FB
CL-FL-S
CL-FL-P
CB-FL-S
CB-FL-P
CL-FB-S
CL-FB-P
CB-FB-S
CB-FB-P
13.1 13.2
13.3 13.4
11.5 11.7
11.9 11.9
11.2 11.3
11.8 11.8
12.3 12.1
14.7 14.7
12.0 11.9
14.3 14.4
12.0 11.5
12.5 11.8
9.8 9.6
10.6 10.3
9.7 9.3
10.2 9.9
10.6 10.0
15.0 14.4
10.2 9.7
14.4 13.8
12.9 12.9
11.6 11.8
10.0 10.1
11.8 11.7
10.0 10.1
11.6 11.5
10.9 10.7
13.6 13.6
10.5 10.4
13.1 13.2
13.9 13.2
14.7 14.3
13.9 14.1
13.2 12.3
13.4 13.2
13.1 12.4
14.2 13.6
15.5 15.2
14.0 13.5
15.3 15.0
13.6 15.1
14.3 15.7
12.2 13.1
12.1 13.3
11.7 12.6
12.1 13.4
13.5 14.2
14.6 15.8
13.2 14.1
14.5 15.5
8 12 16 20
MAE (d)


Forcing response
Linear 11.96
Beta 13.21
Chilling response
Triangular 12.57
Beta 12.28
Chilling–forcing interaction
Sequential 11.72
Parallel 13.14
Process structure
Forcing-only 13.21
Chilling–forcing 12.43
11.0
11.5
12.0
12.5
13.0
13.5
14.0
Overall MAE (d)


Individual site
Median and interquartile range
Fewer than 10 groups
(a)
Median
n
EZ01
No observations
0
EZ02
10.5
41
EZ03
7.9
225
EZ04
7.5
104
EZ05
7.8
94
0
10
20
30
50
80
Site MAE (d)
(b)
Median
n
EZ01
69.0
5
EZ02
12.9
42
EZ03
6.5
198
EZ04
7.0
109
EZ05
8.5
95
0
10
20
30
50
80
Site MAE (d)
(c)
Median
n
EZ01
No observations
0
EZ02
11.8
12
EZ03
14.4
111
EZ04
8.7
95
EZ05
8.1
128
0
10
20
30
50
80
Site MAE (d)
(d)
Median
n
EZ01
No observations
0
EZ02
27.5
1
EZ03
22.2
6
EZ04
8.0
68
EZ05
12.3
80
0
10
20
30
50
80
Site MAE (d)


(a)
Days
FL
FB
CL-FL-S
CL-FL-P
CB-FL-S
CB-FL-P
CL-FB-S
CL-FB-P
CB-FB-S
CB-FB-P
All models
0.86
1.23
0.79
0.96
0.46
0.95
1.08
1.79
0.97
1.47
1.06
0
1
2
Overall MAE reduction (d)
(b)
Sites improved
0.73 d
Budburst
64%
1.61 d
Flowering
66%
0.91 d
Veraison
61%
0.97 d
Maturity/harvest
63%
0
1
2
MAE reduction (d)
BBCH 07
BBCH 65
BBCH 81
BBCH 89
(c)
Site minus grid elevation (m)
≤ -41.7
-41.7 to -0.5
> -0.5
−2
0
2
4
6
MAE reduction (d)
Positive values indicate improvement
(d)
Elevation mismatch
Local elevation variability
Regional sill
0
1
2
3
Overall MAE reduction (d)
Low
Middle
High


(a)
07
65
81
89
FL-ML 4.3 6.2 3.3 5.0
FB-ML 4.6 4.9 3.8 5.5
CL-FL-S-ML 2.7 3.5 2.8 2.8
CL-FL-P-ML 3.4 5.2 2.3 3.2
CB-FL-S-ML 2.3 3.5 2.9 2.9
CB-FL-P-ML 3.0 5.0 2.5 3.5
CL-FB-S-ML 2.8 4.1 3.1 4.1
CL-FB-P-ML 6.9 6.6 5.2 5.5
CB-FB-S-ML 2.5 3.7 3.0 4.0
CB-FB-P-ML 6.4 6.2 5.0 5.6
0
4
8
12
ML gain (d)
(b)
FL-ML
FB-ML
CL-FL-S-ML
CL-FL-P-ML
CB-FL-S-ML
CB-FL-P-ML
CL-FB-S-ML
CL-FB-P-ML
CB-FB-S-ML
CB-FB-P-ML
All models
0
2
4
6
Overall ML gain (d)
(c)
07
65
81
89
EZ01 — no data | 59.9* g = 5 | — no data | — no data
EZ02 7.1 5.1 to 9.0 | 4.1 1.7 to 6.6 | 2.3* g = 8 | 13.8* g = 1
EZ03 2.5 1.9 to 3.1 | 1.4 0.9 to 2.0 | 2.4 1.0 to 3.8 | 10.0* g = 4
EZ04 2.3 1.7 to 2.9 | 4.1 3.1 to 5.2 | 2.9 1.9 to 3.8 | 4.8 2.1 to 7.7
EZ05 7.7 5.8 to 9.6 | 10.6 9.1 to 12.0 | 4.7 3.8 to 5.7 | 3.2 1.8 to 4.4
ML gain and 95% interval (d)
(d)
07
65
81
89
EZ01 — no data | 12.9* n = 5 | — no data | — no data
EZ02 9.3 n = 41 | 10.1 n = 42 | 15.5* n = 12 | 20.3* n = 1
EZ03 7.5 n = 225 | 7.6 n = 198 | 15.2 n = 111 | 15.1* n = 6
EZ04 6.5 n = 104 | 4.9 n = 109 | 8.7 n = 95 | 8.6 n = 68
EZ05 5.3 n = 93 | 4.5 n = 93 | 6.7 n = 128 | 10.8 n = 80
0
5
10
15
20
Hybrid MAE (d)
* Fewer than 10 groups; g = groups
n = sites; all panels use RAW temperature

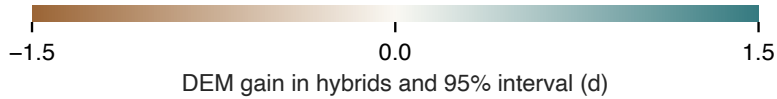


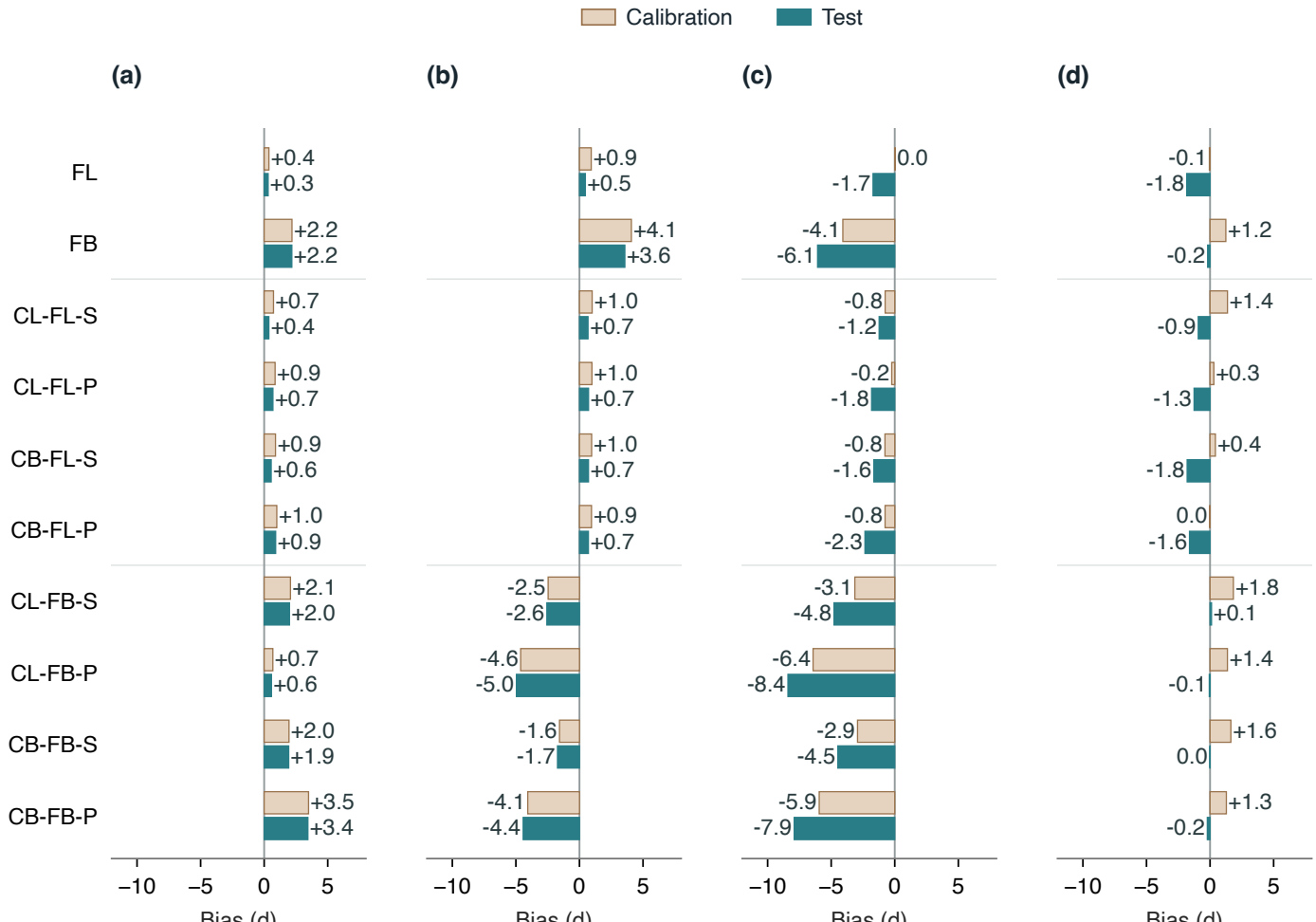
(c)

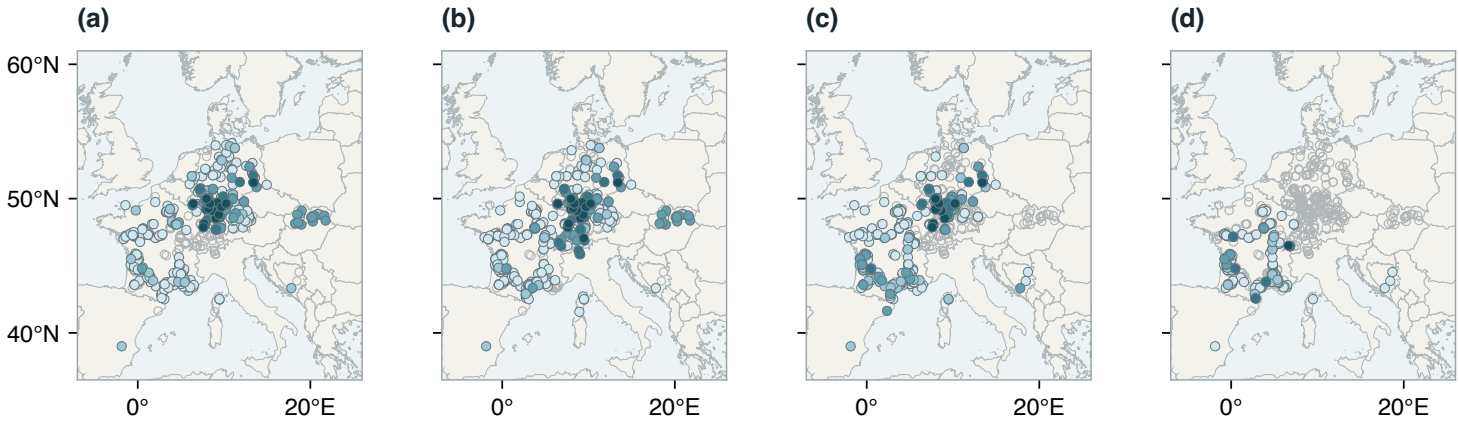
| | 07 | 65 | 81 | 89 |
|---|---|---|---|---|
| EZ01 | —<br>no data | 0.93*<br>g = 5 | —<br>no data | —<br>no data |
| EZ02 | -0.09<br>-0.97 to 0.79 | 0.69<br>-0.23 to 1.70 | 0.23*<br>g = 8 | -4.32*<br>g = 1 |
| EZ03 | 0.25<br>0.02 to 0.49 | 0.00<br>-0.27 to 0.24 | 0.32<br>-0.42 to 1.02 | 2.45*<br>g = 4 |
| EZ04 | 0.42<br>0.04 to 0.76 | 0.32<br>0.03 to 0.59 | 0.17<br>-0.36 to 0.73 | 0.04<br>-0.74 to 0.67 |
| EZ05 | 0.04<br>-0.28 to 0.34 | -0.05<br>-0.38 to 0.27 | -0.05<br>-0.78 to 0.54 | -0.19<br>-0.95 to 0.35 |

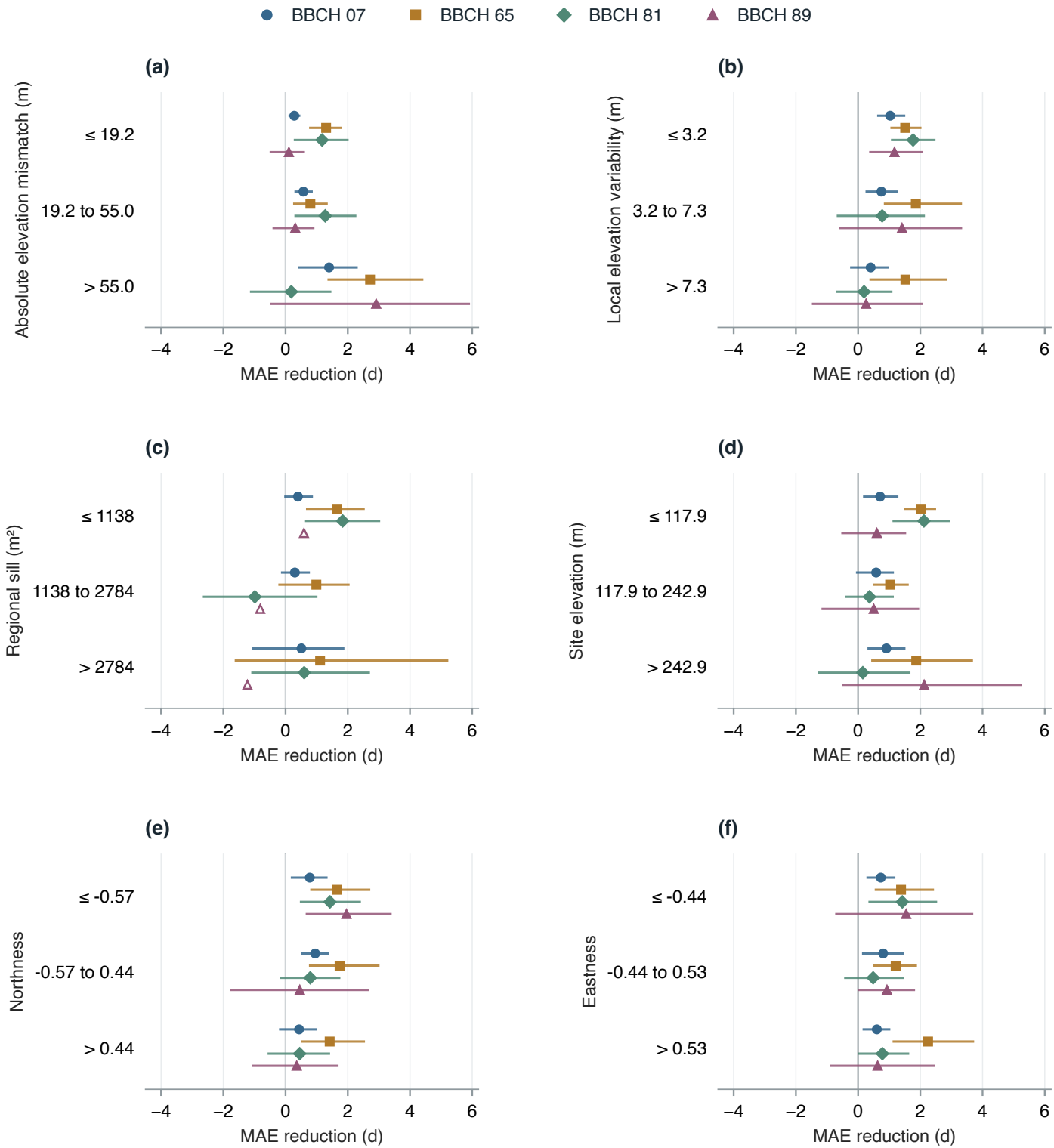
* Fewer than 10 groups

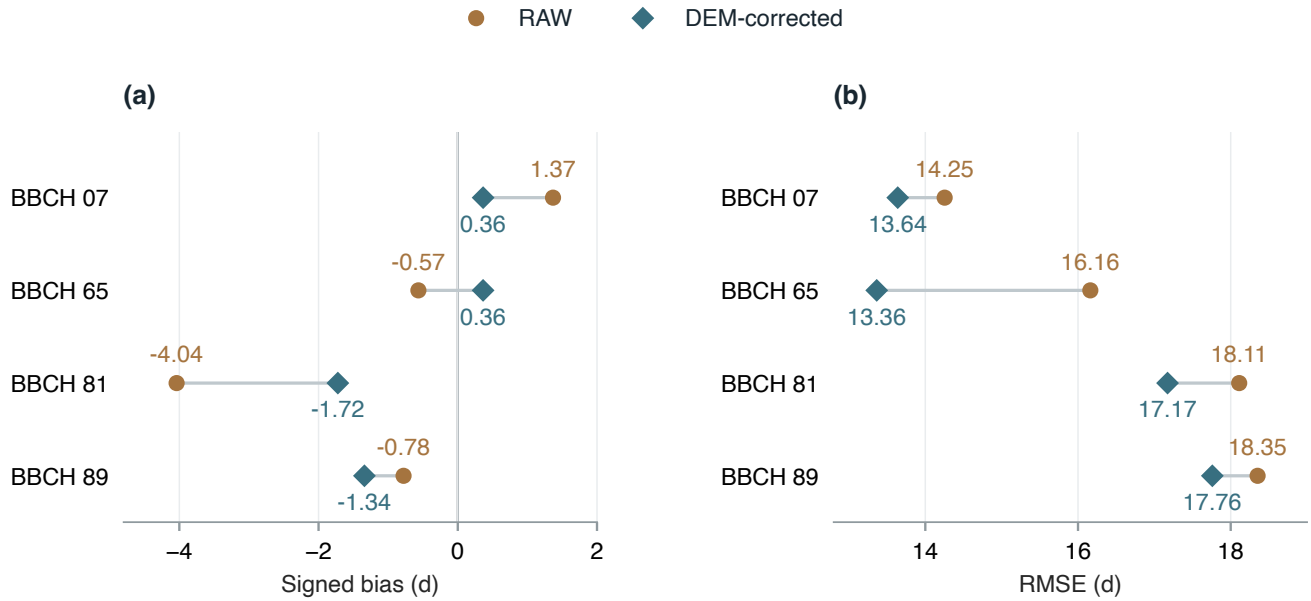
g = groups

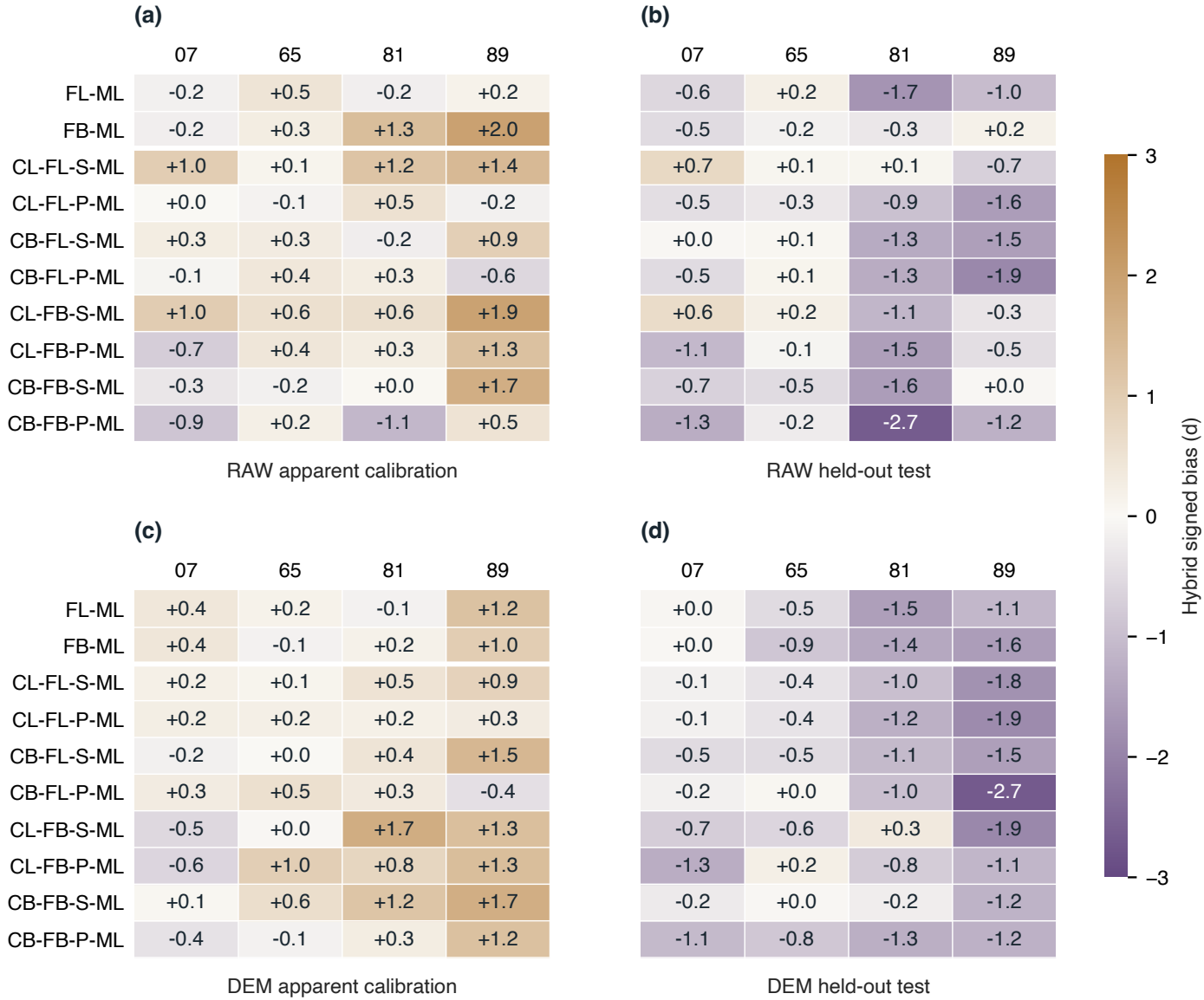



Calibration
Test
(a)
(b)
(c)
(d)
FL
FB
CL-FL-S
CL-FL-P
CB-FL-S
CB-FL-P
CL-FB-S
CL-FB-P
CB-FB-S
CB-FB-P
+0.4
+0.3
+2.2
+2.2
+0.7
+0.4
+0.9
+0.7
+0.9
+0.6
+1.0
+0.9
+2.1
+2.0
+0.7
+0.6
+2.0
+1.9
+3.5
+3.4
+0.9
+0.5
+4.1
+3.6
+1.0
+0.7
+1.0
+0.7
+1.0
+0.7
+0.9
+0.7
-2.5
-2.6
-4.6
-5.0
-1.6
-1.7
-4.1
-4.4
0.0
-1.7
-4.1
-6.1
-0.8
-1.2
-0.2
-1.8
-0.8
-1.6
-0.8
-2.3
-3.1
-4.8
-6.4
-8.4
-2.9
-4.5
-5.9
-7.9
-0.1
-1.8
+1.2
-0.2
+1.4
-0.9
+0.3
-1.3
+0.4
-1.8
0.0
-1.6
+1.8
+0.1
+1.4
-0.1
+1.6
0.0
+1.3
-0.2
−10
−5
0
5
Bias (d)


(a)
(b)
(c)
(d)
60°N
50°N
40°N
0°
20°E
466 sites
5,073 observations
452 sites
5,469 observations
346 sites
3,754 observations
155 sites
850 observations
Observations per site
No observations
1–5
6–10
11–25
26–50
>50


BBCH 07
BBCH 65
BBCH 81
BBCH 89
(a)
Absolute elevation mismatch (m)
≤ 19.2
19.2 to 55.0
> 55.0
(b)
Local elevation variability (m)
≤ 3.2
3.2 to 7.3
> 7.3
(c)
Regional sill (m²)
≤ 1138
1138 to 2784
> 2784
(d)
Site elevation (m)
≤ 117.9
117.9 to 242.9
> 242.9
(e)
Northness
≤ -0.57
-0.57 to 0.44
> 0.44
(f)
Eastness
≤ -0.44
-0.44 to 0.53
> 0.53
−4 −2 0 2 4 6
MAE reduction (d)


RAW
DEM-corrected
(a)
BBCH 07
BBCH 65
BBCH 81
BBCH 89
1.37
0.36
-0.57
0.36
-4.04
-1.72
-0.78
-1.34
−4
−2
0
2
Signed bias (d)
(b)
BBCH 07
BBCH 65
BBCH 81
BBCH 89
14.25
13.64
16.16
13.36
18.11
17.17
18.35
17.76
14
16
18
RMSE (d)


(a)
07 65 81 89
FL-ML -0.2 +0.5 -0.2 +0.2
FB-ML -0.2 +0.3 +1.3 +2.0
CL-FL-S-ML +1.0 +0.1 +1.2 +1.4
CL-FL-P-ML +0.0 -0.1 +0.5 -0.2
CB-FL-S-ML +0.3 +0.3 -0.2 +0.9
CB-FL-P-ML -0.1 +0.4 +0.3 -0.6
CL-FB-S-ML +1.0 +0.6 +0.6 +1.9
CL-FB-P-ML -0.7 +0.4 +0.3 +1.3
CB-FB-S-ML -0.3 -0.2 +0.0 +1.7
CB-FB-P-ML -0.9 +0.2 -1.1 +0.5
RAW apparent calibration
(b)
07 65 81 89
-0.6 +0.2 -1.7 -1.0
-0.5 -0.2 -0.3 +0.2
+0.7 +0.1 +0.1 -0.7
-0.5 -0.3 -0.9 -1.6
+0.0 +0.1 -1.3 -1.5
-0.5 +0.1 -1.3 -1.9
+0.6 +0.2 -1.1 -0.3
-1.1 -0.1 -1.5 -0.5
-0.7 -0.5 -1.6 +0.0
-1.3 -0.2 -2.7 -1.2
RAW held-out test
(c)
07 65 81 89
FL-ML +0.4 +0.2 -0.1 +1.2
FB-ML +0.4 -0.1 +0.2 +1.0
CL-FL-S-ML +0.2 +0.1 +0.5 +0.9
CL-FL-P-ML +0.2 +0.2 +0.2 +0.3
CB-FL-S-ML -0.2 +0.0 +0.4 +1.5
CB-FL-P-ML +0.3 +0.5 +0.3 -0.4
CL-FB-S-ML -0.5 +0.0 +1.7 +1.3
CL-FB-P-ML -0.6 +1.0 +0.8 +1.3
CB-FB-S-ML +0.1 +0.6 +1.2 +1.7
CB-FB-P-ML -0.4 -0.1 +0.3 +1.2
DEM apparent calibration
(d)
07 65 81 89
+0.0 -0.5 -1.5 -1.1
+0.0 -0.9 -1.4 -1.6
-0.1 -0.4 -1.0 -1.8
-0.1 -0.4 -1.2 -1.9
-0.5 -0.5 -1.1 -1.5
-0.2 +0.0 -1.0 -2.7
-0.7 -0.6 +0.3 -1.9
-1.3 +0.2 -0.8 -1.1
-0.2 +0.0 -0.2 -1.2
-1.1 -0.8 -1.3 -1.2
DEM held-out test
Hybrid signed bias (d)
3
2
1
0
−1
−2
−3